SECTION MILL

HYDRAULICALLY ACTUATED CASING SECTION MILL

MILLING

GENERAL DESCRIPTION

HYDRAULICALLY ACTUATED TOOL THAT MILLS A SECTION OUT OF CASING OR TUBING, AND CAN ALSO BE USED TO CUT A SINGLE STRING OF CASING. PUMP PRESSURE ACROSS THE PISTON PUSHES A CAM DOWN AND EXPANDS THE KNIVES AGAINST THE CASING: THE CUT-OUT KNIVES PART THE PIPE FIRST, THEN ALL THE KNIVES TAKE PART IN MILLING. WHEN CIRCULATION STOPS, THE PISTON SPRING RETRACTS THE CAM AND THE KNIVES FOLD BACK INTO THE BODY FOR RETRIEVAL. KNIVES ARE DRESSED WITH TUNGSTEN CARBIDE INSERTS, SUITED TO POORLY CEMENTED, SPLIT OR CORRODED CASING; A SECTION OF UP TO 60 FT IS TYPICALLY MILLED WITH ONE SET OF KNIVES.

TECHNICAL CHARACTERISTICS

Tool series	Casing size	Body Ø	Fishing neck length	Fishing neck Ø	Overall length	Top pin API Reg.	Wt (lb)
3600	4-1/2	3-5/8	18	3-1/8	56	2-3/8	135
4100	5	4-1/8	18	3-1/4	66	2-3/8	175
4500	5-1/2, 6	4-1/2	18	4-1/8	70	2-7/8	220
5500	6-5/8, 7	5-1/2	18	4-3/4	74	3-1/2	350
6100	7-5/8	6-1/8	18	4-3/4	74	3-1/2	368
7200	8-5/8, 9-5/8	7-1/4	18	5-3/4	89	4-1/2	554
8200	9-5/8	8-1/4	18	5-3/4 or 8	87	4-1/2 or 6-5/8	900
9200	10-3/4, 11-3/4	9-1/4	18	5-3/4 or 8	87	4-1/2 or 6-5/8	980
11700	13-3/8, 16	11-1/2	18	8 or 9	90	6-5/8 or 7-5/8	1,725

Dimensions in inches, weights in lb (approximate). Standard API regular pin connections; others on request.

DESIGN FEATURES

- **Staged knife opening:** three knives cut out, then all six mill.
- **Carbide cutting structure:** inserts laid in a brick-work pattern for footage and penetration.
- **Positive cut-out signal:** the Flo-Tel option doubles the flow area once the casing is parted, giving a 200 - 250 psi pressure drop at surface. Standard on series 4100 to 11700; not available on 3600.
- **Self-retracting:** piston spring closes the knives when pumps stop.
- **Jetted top sub:** limits velocity through the piston above 300 gpm.
- **Simple servicing:** replaceable knives, arm-stop stabilizers and Flo-Tel unit.

FLOW RATES (GPM)

Tool series	During cut-out	During milling
3600	110 - 160	110 - 160
4100	80 - 125	110 - 160
4500	80 - 125	150 - 200
5500	80 - 125	200 - 250
6100	80 - 125	200 - 250
7200	80 - 125	200 - 300
8200	80 - 125	300 - 400
9200	80 - 125	350 - 450
11700	350 - 450	350 - 600

OPERATING PARAMETERS

Rotary speed, cut-out	60 rpm to start
Rotary speed, milling	80 - 120 rpm
Weight on mill	4,000 - 8,000 lb
Min. pressure at tool	about 300 psi to cut out
Pipe cutter use	80 - 100 rpm

HOW TO ORDER

Please specify: 1. tool series 2. size and weight of casing to be milled 3. top pin connection.